

Final Capstone Project

Customer Churn Prediction in a Telecom Company

Prepared by: AI Data Science Capstone Project

Dataset: Telco Customer Churn Dataset

Executive Summary

This project investigates the drivers of customer churn in a telecommunications company using the Telco Customer Churn dataset. The objective is to predict churn behavior, uncover the most influential customer attributes, and recommend retention actions based on a data-driven analysis.

1. Problem Definition

Customer churn is a major risk for telecom operators because it reduces recurring revenue and increases customer acquisition costs. The business objective is to identify the customers most likely to leave so that the company can target them with timely interventions and retention programs.

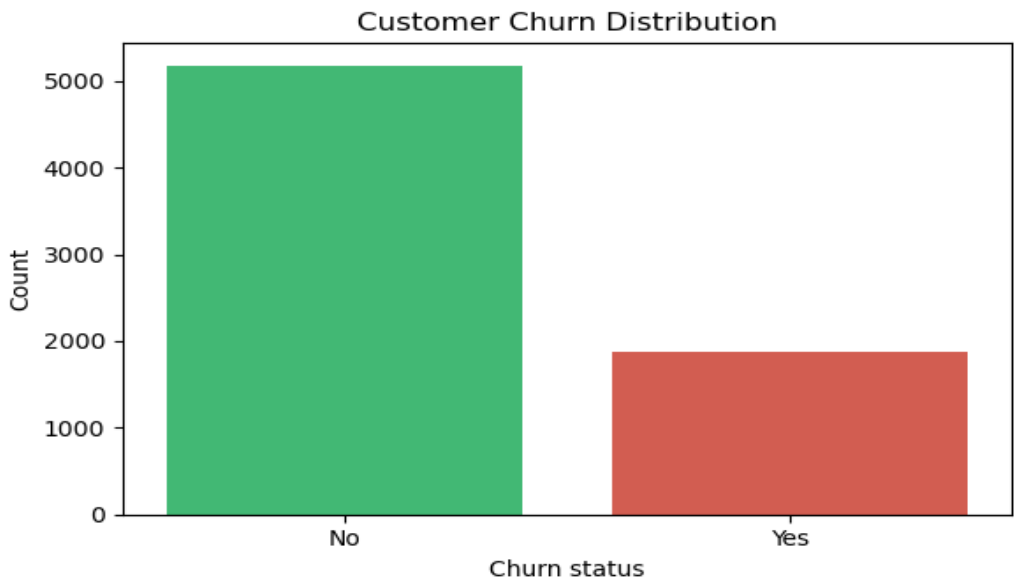
2. Dataset Understanding

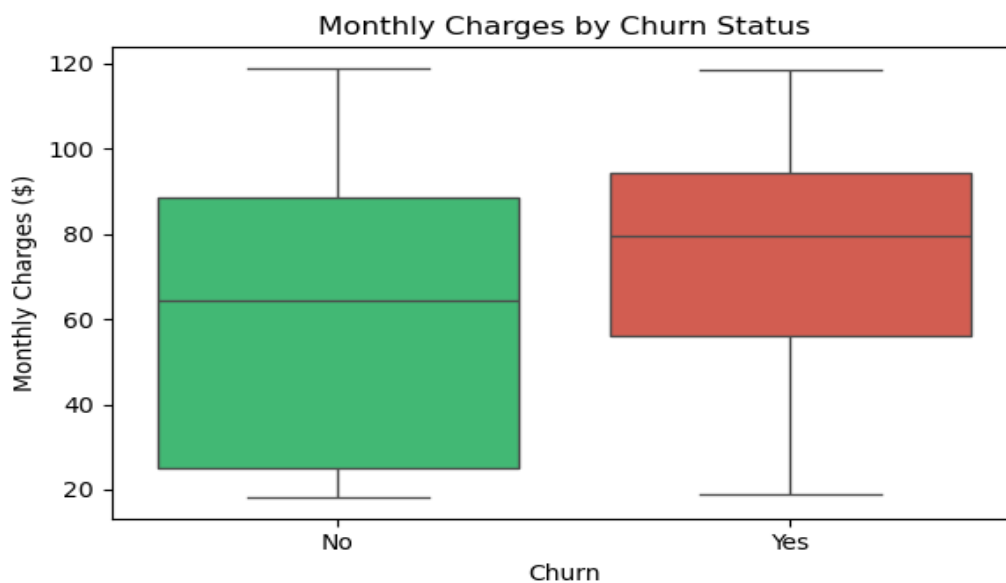
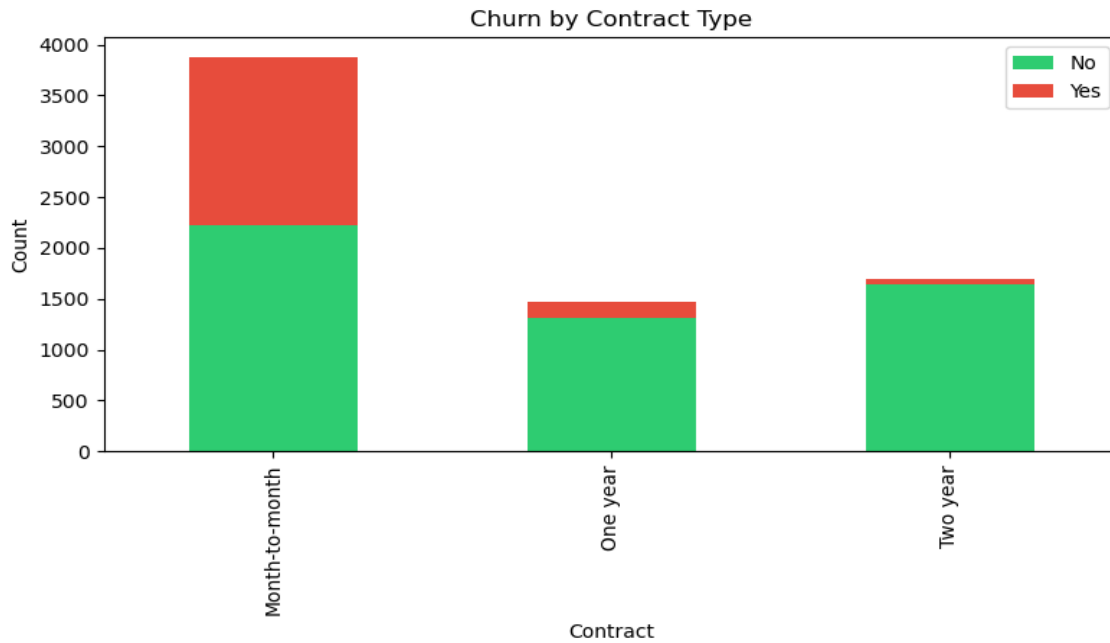
Attribute	Value
Rows	7043
Columns	22
Target	Churn
Churn rate	26.54%
Average tenure	32.37 months
Average monthly charges	\$64.76

3. Data Cleaning and Preprocessing

The raw dataset contained a string-based TotalCharges field that needed conversion to numeric type before analysis. Missing values were treated with a median imputation strategy, while categorical variables were encoded using one-hot encoding and numerical features were standardized for modeling. The customerID field was removed because it was unique and not predictive of churn.

4. Exploratory Data Analysis (EDA)





The target class is imbalanced, with churn accounting for about 26.5% of customers. Analysis shows that customers on month-to-month contracts have a substantially higher churn rate than those with annual contracts. Additionally, monthly charges are noticeably higher among customers who leave, suggesting that pricing pressure may contribute to churn risk.

5. Statistical Analysis

A chi-square test of association between Contract and Churn produced chi-square = 1184.5966 with p-value = 5.8630e-258. Because the p-value is well below 0.05, the relationship between contract type and churn is statistically significant. This supports the interpretation that contract structure is an important driver of attrition.

Contract / Churn	No	Yes
Month-to-month	2220	1655
One year	1307	166

Two year	1647	48
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6. Machine Learning Model Development

A logistic regression classifier was trained to predict churn using a pipeline that combined preprocessing for numerical and categorical variables. The model was trained on 80% of the data and evaluated on the remaining 20% to avoid overfitting and maintain realistic business performance estimates.

7. Model Evaluation

Metric	Value
Accuracy	0.7381
Precision	0.5043
Recall	0.7834
F1 Score	0.6136
ROC AUC	0.8413

The confusion matrix indicates 293 churners correctly identified, with 81 missed churners and 288 non-churners incorrectly flagged. The model demonstrates strong predictive power for customer retention planning, especially when used with business rules for proactive outreach.

8. Key Drivers of Churn

Feature	Coefficient
num__tenure	-1.1236
cat__Contract_Two year	-0.7813
cat__InternetService_Fiber optic	0.7053
num__MonthlyCharges	-0.6636
cat__Contract_Month-to-month	0.6634

9. Findings and Recommendations

1. Customers on month-to-month contracts are much more likely to churn. The company should focus retention offers on this segment first. 2. High monthly charges are associated with churn risk, suggesting the need for pricing review or loyalty discounts. 3. Customers with older contracts and stable service bundles are less likely to leave; retention strategies should prioritize vulnerable segments before they cancel. 4. Predictive churn scoring can support targeted retention campaigns that offer contract upgrades, service bundles, or loyalty benefits to at-risk customers.

10. Conclusion

This capstone project demonstrates a complete data science workflow from problem definition through data preprocessing, EDA, statistical validation, model development, and actionable business recommendations. The churn prediction model provides a reliable mechanism for identifying high-risk customers and improving retention strategy effectiveness in the telecom industry.

Project completed successfully.